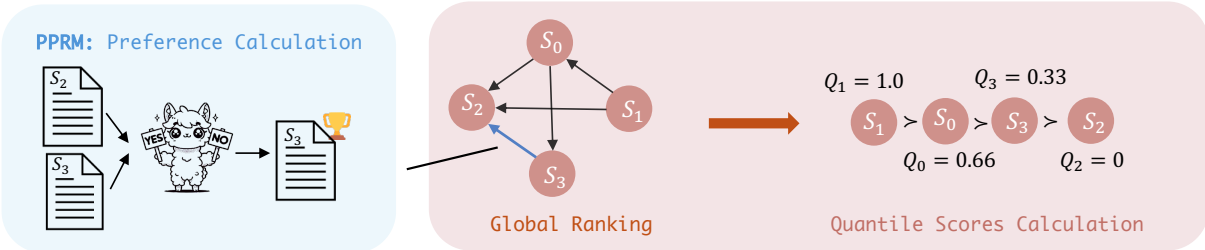

Figure 2: Preference prediction process of PPRM and global quantile score based on Enhanced Borda Count method.

of s_i as a discounted sum of its child nodes' Q values: $Q(s_i) = (1-\gamma)Q(s_i) + \gamma Q(s')$. The discount factor γ represents the importance of future rewards. This iterative update mechanism ensures that the values of parent nodes are progressively refined, enhancing the guidance for future selections.

Additionally, to control the growth of the search tree, the SR-MCTS method restricts the maximum number of rollouts N_{max} . The search process terminates when the restriction is reached, imposing limits on the unbounded expansion of the tree. The overarching objective of SR-MCTS is to maximize the expected highest Q value of all existing nodes S, guiding us towards the most desirable outcome s^*, ensuring that the search process efficiently converges to high-quality solutions.

2.3 Pairwise Preference Reward Model

Reliable evaluation of different solutions plays a crucial role in mathematical problem-solving tasks as it leads to better estimation of Q-values, thereby offering better guidance. Existing reward models typically evaluate solutions by giving absolute scores, such as process reward model (PRM Lightman et al., 2023) and outcome reward model (ORM Yu et al., 2023). However, the score-based reward models may fall short in leveraging the instruction-following capability of LLMs or effectively handling the variations in scoring standards, especially when the differences between solutions are subtle. To address this issue, we propose the Pairwise Preference Reward Model (PPRM), which leverages a comprehensive preference dataset incorporating substantial samples from both PRM and ORM approaches (Toshniwal et al., 2024; Lightman et al., 2023) to learn preference relationships among mathematical solutions.

For two solutions (a_1 and a_2) to a given mathematical problem, we use $a_1 \succ a_2$ to represent the situation where a_1 is preferred over a_2. PPRM predicts their relative quality using a pairwise partial ordering, represented by the following probability formula:

$$P(a_1 \succ a_2 \mid \phi) = \frac{e^{\phi(a_1)}}{e^{\phi(a_1)} + e^{\phi(a_2)}} \tag{4}$$

, where $P(a_1 \succ a_2 \mid \phi)$ denotes the probability of a partial ordering relation between solution a_1 and a_2, with ϕ representing the parameters of the reward model. In our method, $a_1 \succ a_2$ are represented by tokens of an LLM, and $P(a_1 \succ a_2 \mid \phi)$ is estimated using the logits value of tokens calculated by the LLM.

Then, inspired by advancements in Language Interface Fine-Tuning (LIFT Dinh et al., 2022), we frame the training process of PPRM as a question-answering task to leverage the instruction-following capabilities of LLMs. The model is tasked with answering the question, "For Question Q, is solution a_1 better than solution a_2?" as shown in Figure 2. To form a robust training objective, the predicted token labels $\hat{y}$ ('Yes' or 'No') are evaluated with ground truth label y using the indicator function $\mathbf{I}$:

$$\mathbf{I}(\hat{y}, y) = \begin{cases} 1, & \text{if } \hat{y} = y \\ 0, & \text{if } \hat{y} \neq y \end{cases} \tag{5}$$

Finally, a pairwise preference dataset $\mathcal{D}$ that contains millions of mathematical problem-solving solution pairs is converted into a dataset $\mathcal{D}'$ suitable for a question-answering task. We employ RLHF techniques to train the model to improve its performance in the partial-order prediction question-answering task. Subsequently, the Direct Preference Optimization (DPO Rafailov et al., 2024) method is utilized to find the optimal P_ϕ by maximizing the objective $\arg\max_\phi \mathbb{E}_P[\mathbf{I}(\hat{y}, y)]$. Please refer to Appendix C for details about training and inference of PPRM.

2.4 Enhanced Borda Count Method

Although PPRM allows us to directly compare the quality of two solutions, we still need to convert these local preferences into a cohesive global ranking to gain a comprehensive evaluation for the answers. This conversion process can be formalized as the global optimal ranking aggregation (GORA) problem related to Learning to Rank (LtR) methods. Further, we propose the Enhanced Borda Count (EBC) method based on the transitivity assumption of mathematical problem solutions, which integrates the naive Borda Count algorithm with a transitive closure of preferences calculated by the Floyd-Warshall (Warshall, 1962) algorithm. For formalized discussion, please refer to Appendix H.

Local preference calculation. First, the PPRM generates a preference matrix $\mathbf{M} \in \mathbb{R}^{n \times n}$ for all n problem solutions, where $\mathbf{M}_{i,j} = 1$ indicates that solution a_i is superior to solution a_j, and $\mathbf{M}_{i,j} = 0$ otherwise. This process can be represented as:

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } P(a_i \succ a_j) \geq 0.5 \\ 0, & \text{if } P(a_i \succ a_j) < 0.5 \end{cases} \tag{6}$$

As shown in Figure 2, this matrix can be viewed as an adjacency matrix of a directed graph $\mathbf{G} = (\mathbf{V}, \mathbf{E})$, where each solution a_i corresponds to a vertex v_i, and an edge $e = (v_i, v_j)$ exists if $\mathbf{M}_{i,j} = 1$, indicating that solution a_i is preferred over a_j.

Transitive closure. To simplify the problem, we adopt the assumption of transitivity for mathematical solutions, that is, if $v_i \succ v_k$ and $v_k \succ v_j$, then $v_i \succ v_j$. Under this assumption, the transitive closure $\mathbf{C}$ of a preference matrix can be computed through Floyd-Warshall algorithm, e.g., if $\mathbf{M}_{i,k} = 1$ and $\mathbf{M}_{k,j} = 1$, then $\mathbf{M}_{i,j} = 1$.

Borda count-based global ranking. Next, based on the transitive closure matrix $\mathbf{C}$, we apply the Enhanced Borda Count method for global ranking. The Enhanced Borda Count determines the ranking of each node by calculating its out-degree, which corresponds to the number of nodes it defeats. For each node v_i, the $\text{Borda}(v_i)$ is defined as $\sum_{j=1}^{n} \mathbf{C}_{i,j}$, like the ranked node listed in Figure 2.

Nodes with higher Borda counts are ranked higher. However, in practice, cyclic preferences can cause efficiency issues with the naive Borda Count method. To further refine the ranking, we devise a re-ranking stage, where the logits generated by the PPRM are used for soft comparison among nodes with equal Borda counts. Specifically, for two nodes v_i and v_j with equal Borda counts, the soft comparison rule can be denoted as $v_i \succ v_j \iff P(a_i \succ a_j) > P(a_j \succ a_i)$. This process ensures that the ranking remains consistent and reasonable even in the presence of cycles or local ambiguities.

Global quantile score of solutions. Finally, the ranking is converted into a global quantile score Q_g for each solution v is $Q_g(v) = 1 - \frac{\text{rank}(v)-1}{|V|-1}$, where $\text{rank}(v)$ is the position of v in the ranking based on Borda counts, and $|V|$ is the total number of nodes. To reflect local advantages in the search tree structure, the local win rate $Q_l(v)$ for a node v is calculated in $\mathbf{C}$ with its children nodes Children_v as follows:

$$Q_l[v] = \frac{\sum_{u \in \text{Children}[v]} C[u, v]}{|\text{Children}[v]|} \tag{7}$$

Finally, score $Q(v)$ for a solution is a weighted sum of local win rate $Q_l(v)$ and global quantile score Q_g.

3 Evaluation

3.1 Experiment Settings

Settings. To better evaluate the effectiveness of our approach, we select LLaMA-3.1-8B-Instruct model (Meta, 2024b) as the base model for SR-MCTS, **without any additional training**. We also train a Gemma2-2B-Instruct model (Google, 2024) as PPRM to provide reward signals during the search process. We develop the `Berry-Tree` inference framework to ensure robust and efficient inference, which supports advanced features, including fault tolerance, checkpoint recovery, multi-query concurrency, and automatic multi-server load balancing. Hyper-parameter settings are detailed in Appendix A.

Grading. We evaluate algorithm-generated answers using the correctness evaluation standards as in Lightman et al. (2023), focusing on format adherence and content accuracy. The model is provided with a prompt specifying the expected answer format. We score answers as consistent if they exactly match the ground truth, closely approximate it numerically, or are equivalent in symbolic form. To ensure a comprehensive and rigorous evaluation, we adopt **major@k** (Kuncheva, 2014) and **rm@k** (Yang et al., 2024c), which can be unified as the solved rate of problems (Lightman et al., 2023;

Table 1: Performance comparison of models across benchmarks of different difficulties, as represented by GaoKao2023En (Liao et al., 2024), College Math (Tang et al., 2024), and OlympiadBench (He et al., 2024), which range from high school to Olympiad levels. Scores denoted with subscripted notations, such as `maj@8`, represent specific metrics, with **major@8** as an example. Scores without subscripted notations reflect the model's greedy performance evaluated in a zero-shot CoT manner.

Model \ Benchmark	GSM8K	MATH	GaoKao2023En	OlympiadBench	College Math	MMLU STEM
Qwen2-7B-Instruct (Yang et al., 2024a)	85.7	52.9	36.4	21.3	24.5	68.2
Meta-Llama-3.1-8B-Instruct (Meta, 2024b)	76.6	47.2	30.1	15.4	33.8	60.5
Qwen2-72B-Instruct (Yang et al., 2024a)	93.2	69.0	58.7	33.2	43.2	84.4
Meta-Llama-3.1-70B-Instruct (Meta, 2024a)	94.1	65.7	54.0	27.7	42.5	80.4
DeepSeekMath-7B-RL (Shao et al., 2024)	88.2	52.4	43.6	19.0	37.5	64.8
Internlm2-math-plus-7b (Ying et al., 2024)	84.0	54.4	50.1	18.8	36.2	55.2
Mathstral-7B-v0.1 (Mistral AI, 2024)	84.9	56.6	46.0	21.5	33.7	64.0
NuminaMath-7B-CoT (Beeching et al., 2024b)	75.4	55.2	47.5	19.9	36.9	60.8
Qwen2-Math-7B-Instruct (Yang et al., 2024a)	89.9	75.1	62.1	38.2	45.9	63.8
NuminaMath-72B-CoT (Beeching et al., 2024a)	90.8	66.7	58.4	32.6	39.7	64.5
Qwen2-Math-72B-Instruct (Yang et al., 2024a)	96.7	84.0	68.3	43.0	47.9	79.9
Meta-Llama-3.1-8B-Instruct (Meta, 2024b)	$89.8_{\texttt{maj@8}}$	$54.8_{\texttt{maj@8}}$	$36.4_{\texttt{maj@8}}$	$24.8_{\texttt{maj@8}}$	$36.4_{\texttt{maj@8}}$	$68.3_{\texttt{maj@8}}$
+ LLaMA-Berry (Ours)@8	$94.9_{\texttt{rm@8}}$	$69.4_{\texttt{rm@8}}$	$61.6_{\texttt{rm@8}}$	$47.2_{\texttt{rm@8}}$	$63.7_{\texttt{rm@8}}$	$82.9_{\texttt{rm@8}}$
+LLaMA-Berry (Ours)@16	$96.1_{\texttt{rm@16}}$	$75.3_{\texttt{rm@16}}$	$68.6_{\texttt{rm@16}}$	$55.1_{\texttt{rm@16}}$	$68.9_{\texttt{rm@16}}$	$88.3_{\texttt{rm@16}}$

Luo et al., 2024a). The evaluation methods and metric details are further outlined in Appendix B.

3.2 Benchmarks

General mathematical reasoning benchmarks. We summarize the results on general mathematical reasoning benchmarks in Table 1, which indicates that our method significantly boosts the base model's performance. The results consistently demonstrate improvements across various levels of difficulty. Specifically, the solved rate of problems in 16 rollouts of Meta-Llama-3.1-8B-Instruct (Meta, 2024b) has been improved by more than 35% on three benchmarks. Qwen2-Math-72B-Instruct (Yang et al., 2024b) exhibits the strongest mathematical reasoning capability among the competing methods, while our `LLaMA-Berry`, built on a base model with only 8B parameters, exceeds it in the solved rate of problems on four benchmarks. In particular, `LLaMA-Berry` reaches 55.1% on OlympiadBench and 68.9% on College Math, surpassing it by 11.9% and 21%, respectively.

Cutting-edge mathematical Olympiad benchmarks. In Table 2, we compare the performance of `LLaMA-Berry` with other leading models on Olympic-level benchmarks. The results demonstrate that `LLaMA-Berry` is highly competitive on these benchmarks, demonstrating its capability in complex reasoning. Notably, on the most challenging AIME2024 benchmark, our method boosts the base model's solving rate from 2/30 to 8/30, surpassing typical open-source models and commercial closed-source models, except for the OpenAI o1 series (OpenAI, 2024).

In addition to excelling in mathematical reasoning, our approach also excels across various science and engineering domains. For example, it achieves top performance on benchmarks such as MMLU STEM (Hendrycks et al., 2021a) in Table 1 and GPQA diamond (Rein et al., 2024) in Table 2. This demonstrates the method's robustness and versatility, enabling it to tackle a wide range of technical challenges and highlighting its potential for broader applications in both research and practical scenarios.

Table 2: Performance comparison across multiple olympiad benchmarks, including AIME24 (MAA, 2024), AMC23 (MAA, 2023), Math Odyssey (Fang et al., 2024), and GPQA Diamond (Rein et al., 2024).

Model	Benchmarks			
	AIME24	AMC23	Math Odyssey	$\text{GPQA}_{\texttt{Diamond}}$
Claude 3 Opus	6.7	42.0	40.6	50.4
GPT 4 Turbo	3.3	–	47.0	38.8
GPT 4o	13.4	–	–	56.1
OpenAI o1 Preview	56.7	–	–	78.3
OpenAI o1	83.3	–	–	78.0
Gemini 1.5 Pro	6.7	–	45.0	–
Gemini Math-Specialized 1.5 Pro	23.3	–	55.8	–
Meta-LLaMA-3.1-8B-Instruct	6.7	15.7	41.7	30.4
Qwen2-Math-7B-Instruct	13.3	62.5	–	–
NuminaMath-72B CoT	3.3	52.5	–	–
Qwen2-Math-72B-Instruct	20.0	60.0	–	–
Meta-Llama-3.1-8B-Instruct	$13.3_{\texttt{maj@8}}$	$22.9_{\texttt{maj@8}}$	$44.2_{\texttt{maj@8}}$	$39.4_{\texttt{maj@8}}$
+ LLaMA-Berry (Ours)@8	$16.7_{\texttt{rm@8}}$	$48.2_{\texttt{rm@8}}$	$60.4_{\texttt{rm@8}}$	$77.3_{\texttt{rm@8}}$
+LLaMA-Berry (Ours)@16	$26.7_{\texttt{rm@16}}$	$54.2_{\texttt{rm@16}}$	$65.0_{\texttt{rm@16}}$	$92.4_{\texttt{rm@16}}$

Comparison with other tree-based or CoT methods. We compare our algorithm with other tree-based reasoning methods and CoT-based methods on GSM8K (Cobbe et al., 2021), GSMHard (Gao et al., 2022), and MATH500 (Lightman et al., 2023) which is a representative and highly challenging 10% subset of MATH (Hendrycks et al., 2021b)